J-Math evaluation of The Euler-Mascheroni Constant

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⋆ Let $f(K) = \frac{(-1)^{(K+1)}}{(K+1)n^{(K+1)}}$ [function]

⋆ Let partial sum of $f(K)$ be
defined as a function of n

$$S(n) = \sum_{K=1}^{n} f(K)$$

Extend the sum from $K = 1$ to $K = 0$

and substract the extra term

$$S(n) = \left(\sum_{K=0}^{n} f(K) \right) - f(0)$$

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find $S(0)$

$$S(0) = \left(\sum_{K=0}^{0} f(K) \right) - f(0) = \cancel{f(0)} - \cancel{f(0)}$$

$$\boxed{S(0) = 0} \quad ---(1)$$

find $S(1)$

$$S(1) = \left(\sum_{K=0}^{1} f(K) \right) - f(0)$$

$$S(1) = (\cancel{f(0)} + f(1)) - \cancel{f(0)}$$

$$\boxed{S(1) = f(1)} \quad ---(2)$$

Using UGF, we get

$$S(J) = S(0) + [S(1) - S(0)]J$$

sub from Eq (1) & (2)

, we get

$$S(J) = 0 + (f(1) - 0)J$$

$$\boxed{S(J) = f(1)J}$$

since $f(1) = \frac{(-1)^{\overset{\text{OR } 1+1}{(1+1)}}}{(1+1)n^{(1+1)}} = \frac{(-1)^2}{2n^2} = \frac{1}{2n^2}$

we get

$$\boxed{S(J) = \left(\frac{1}{2n^2}\right)J} \quad - - - (3)$$

⋆ Now the Euler-Mascheroni Constant

$$\gamma = \sum_{n=1}^{\infty}\left[\frac{1}{2n^2} - \frac{1}{3n^3} + \frac{1}{4n^4} - \frac{1}{5n^5} - \dots\right]$$

$$\hookrightarrow S(J) = \left(\frac{1}{2n^2}\right)J$$

$$\gamma = \sum_{n=1}^{\infty}\frac{1}{2n^2}J$$

since $\frac{1}{2}, J$ are constant
, we take them out ~~of~~

$$\gamma = \frac{1}{2}J\sum_{n=1}^{\infty}\frac{1}{n^2}$$

since $\sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}$ [Basel Problem]

we get

$$\gamma = \frac{1}{2}J\frac{\pi^2}{6}$$

$$\boxed{\gamma = \frac{\pi^2 J}{12}} \quad - - - (4)$$

OR

$$\boxed{\gamma = \frac{\zeta(2)}{2}J}$$

$\boxed{\text{Destinally Equal to } \gamma}$

since $(-5)! = \frac{(-1)!}{24}$ or $\frac{V_0}{24}$ or $\frac{J}{24}$ $\quad ---(5)$

we get ~~from~~ Eq (4)

Multiply b/s by (1/2), we get

$$\frac{1}{2} \times \gamma = \frac{1}{2} \times \frac{\pi^2}{12} J$$

$$\frac{\gamma}{2} = \frac{\pi^2 J}{24}$$

from Eq (5) , we get

$$\frac{\gamma}{2} = \pi^2 (-5)!$$

$$\boxed{\gamma = 2\pi^2 (-5)!}$$

$\hookrightarrow$ Destinally Equal to γ.